AICR · PARTIAL SOLUTION

Extraction for Split Unpredictable Sources, and Decomposition on a Region

Resolves. *Decomposition for Split Unpredictable Sources* (see `main.tex` in this folder). **Partially.** Three unconditional extraction bounds are proved outright, one of them free of the query budget altogether and tight in its dependence on M; the conjecture itself is proved, for *every* observer, on a region of P, q and M, and remains open in general.

Reader's guide

What is proved. Write $\kappa^{\mathrm{na}}(q)$ for the extraction advantage restricted to observers whose query positions, for each fixed coin string, do not depend on the challenge value, and $\mu(s) := \min\bigl(s\delta, 2(s\delta^2)^{1/3}, 1\bigr)$. Put $\Lambda := \min\bigl\{5\sqrt{\sigma'\delta},\ \sqrt{2\delta\ln(eN)} + 4\delta\sqrt{\sigma'}\bigr\}$, the second arm being the smaller once $\delta \le 0.85$.

(A) $\kappa^{\mathrm{na}}(q) \le \Lambda + \mu(q)$ for every q, and $\kappa(0) \le \Lambda$ against *all* observers. Unconditional (Theorem 1). This settles the statement's assertion that queries whose positions ignore the challenge value are almost free at every budget: their cost is the additive $\mu(q) \le q\delta$, with no multiplier on the leading term and no dependence on M.

(B) $\kappa(q) \le \Lambda + \mu(\min(qM, N^2))$ for every q and every observer. Unconditional (Theorem 2).

(D) $\kappa(q) \le 5\sqrt{\sigma'\delta} + 2\delta\sqrt{M}$ for every q and every observer — indeed for observers of *unbounded* query count. Unconditional (Theorem 3). Its display mentions neither q nor σ, and the collected form above is free of q and retains σ only inside σ'; for every $q \ge 1$ and $M \ge 4$ it is the smaller of (B) and (D) wherever either says anything at all (Corollary 3).

(C) An explicit family, chosen before q, indexed by the oracle and the leakage, and consistent with the oracle, with $\mathsf{Adv}_{\mathcal{Y},D} \le \varepsilon(D) + 2(\sigma' + \log\gamma^{-1})q/P + \gamma + P\delta + q\delta$ (Theorem 4).

(E) Hence the conjecture, with the absolute constants $c = 2$, $C = 8$, for every $P \le \sqrt{\sigma'/\delta}$ and every q with $q^{+}\delta \le \sigma'$, and for *every* observer once $M \le \sigma' q^{+}/(4\delta)$ (Theorem 5). On that region no hypothesis whatever on how the observer's probes depend on the challenge value is needed.

(F) The term $2\delta\sqrt{M}$ of (D) is tight to within a factor 12, already for observers whose query positions ignore the challenge (Proposition 2). No bound free of q can do better in M.

What is not proved. The conjecture in full. Two obstructions survive, stated as mechanisms in §8: challenge-steered probes at M *beyond* $\sigma' q^{+}/(4\delta)$, and large P with growing q. The statement's own remark already names the second as a separate problem.

Where to be most sceptical. Two rounds of blind review found two serious defects, both since repaired, and both of the same species — a claim left *underdetermined* rather than false. They were at Theorem 5's first proof line (which quantity the cited corollary actually bounds) and in Definition 2 (whether challenge resolution is a condition on the law of the query positions or on each coin string). Those two places are where a third error, if there is one, is most likely to be. Since then the document has grown by Lemma 4, Lemma 6, Theorem 3, branch 1 of Theorem 5 and Proposition 2, none of which has been reviewed at all. The review status is recorded honestly in §10.

Notation is that of the statement document and is not repeated: $\mathsf{Fun} = \{f : [N] \times [N] \to [M]\}$, $H \leftarrow \mathsf{Fun}$, split δ-unpredictable sources (S_1, S_2) with leakage bounded by σ_1, σ_2, $\sigma = \sigma_1 + \sigma_2$, $\sigma' = \sigma + 2\log N$, $q^+ = q + 1$, and the experiments $\mathsf{Real}, \mathsf{Dec}, \mathsf{Real}_0, \mathsf{Dec}_0$ with $\mathsf{Adv}_{\mathcal{Y},D}$ and $\kappa(q)$. Throughout $N \geq 2$ and $M \geq 2$; logarithms are base two. Write $\mathcal{Z} := \{0,1\}^{\leq \sigma_1} \times \{0,1\}^{\leq \sigma_2}$, so $|\mathcal{Z}| = (2^{\sigma_1+1} - 1)(2^{\sigma_2+1} - 1) < 2^{\sigma+2}$.

1 The two structural facts

Everything below rests on two observations, which together turn the conjecture into a statement about the discrepancy of a random matrix on combinatorial rectangles.

Lemma 1 (derandomisation). *Fix one of the two paired experiments,* $(\mathsf{Real}, \mathsf{Real}_0)$ *or* $(\mathsf{Real}, \mathsf{Dec})$. *For every q and every q-query challenge-oblivious D there is a deterministic such D' with* (i) *the advantage* in that pair *no larger for D than for D' — that is,* $\left|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\right|$ *in the first case and* $\mathsf{Adv}_{\mathcal{Y},D} \leq \mathsf{Adv}_{\mathcal{Y},D'}$ *in the second — and* (ii) *D' of the same challenge resolution as D (Definition 2). Consequently the suprema defining $\kappa(q)$ and $\kappa^{\mathrm{na}}(q)$ are both attained on deterministic observers, the latter on deterministic observers of resolution* 1.

The maximiser is chosen per pair and the two need not coincide: the lemma does *not* supply one D' dominating both experiments, and nothing here asks it to. Every appeal to it — in Theorems 1, 2 and 3 — is to the pair $(\mathsf{Real}, \mathsf{Real}_0)$, and Theorem 4 is stated for a fixed D with no supremum taken, so $\mathsf{Adv}_{\mathcal{Y},D}$ is never bounded by passing to a derandomised observer.

Proof. Let ρ be D's coins and D_ρ the deterministic strategy it then runs. All of $\mathsf{Fun}, [N], [M], \mathcal{Z}$ are finite and D makes at most q queries, so there are finitely many deterministic q-query challenge-oblivious strategies. Each quantity at issue is $|\mathbb{E}_\rho[\alpha(\rho)]|$ with $\alpha(\rho) = \Pr[\mathsf{E}_1 = 1 \mid \rho] - \Pr[\mathsf{E}_0 = 1 \mid \rho]$, the coins being independent of every other draw and the paired experiments differing in no other component; so it is at most $\max_\rho |\alpha(\rho)|$, attained at some ρ^* depending on the pair fixed at the outset. Take $D' := D_{\rho^*}$, which is q-query and challenge-oblivious. It inherits D's challenge resolution: the definition quantifies over every coin string, so if $D_\rho^f(v, \zeta)$ and $D_\rho^f(v', \zeta)$ issue the same query positions for every ρ whenever $h(v) = h(v')$, they do so in particular at ρ^*, which is what D' runs. The argument is identical for whichever paired experiment was fixed, and is run separately for each. ◻

Once D is deterministic and its oracle is fixed to f, its output is a function of its challenge input alone; write $\theta : [M] \to \{0,1\}$ for that function and $p_\theta := |\theta^{-1}(1)|/M$. That the test is *named by short data* is the crux of Lemma 5.

Lemma 2 (posterior product structure). *For every (f,ζ) in the support of $(H,\mathbf{z})$ the conditional law of $\mathbf{x}$ is the product $\Pi_{f,\zeta} = \pi^1_{f,\zeta_1} \otimes \pi^2_{f,\zeta_2}$, where $\pi^i_{f,\zeta_i}(u) = \Pr\left[x_i = u \mid H = f,\ z_i = \zeta_i\right]$. Writing $m_i := \|\pi^i_{f,\zeta_i}\|_\infty$:*

1. $\max_{\mathbf{u}\in[N]^2} \Pi_{f,\zeta}(\mathbf{u}) = m_1 m_2$*;*
2. $\mathbb{E}[m_i] \le \delta$ *for* $i = 1,2$*;*
3. $\mathbb{E}[m_1 m_2] \le \delta$*;*
4. $\mathbb{E}\left[\sqrt{m_1 m_2}\right] \le \delta$*;*
5. $\Pr[m_1 m_2 > \vartheta] \le \delta/\sqrt{\vartheta}$ *for every* $\vartheta \in (0,1]$.

Proof. The sources run on independent private coins and on the same H, so conditioned on $H = f$ their output pairs are independent: $\Pr\left[x_1 = u_1, z_1 = \zeta_1, x_2 = u_2, z_2 = \zeta_2 \mid f\right] = \prod_i \Pr\left[x_i = u_i, z_i = \zeta_i \mid f\right]$. Summing over u_1, u_2 gives $\Pr\left[\mathbf{z} = \zeta \mid f\right] = \prod_i \Pr\left[z_i = \zeta_i \mid f\right] > 0$, and dividing gives the product form. Item 1 is the fact that the maximum of a product measure is the product of the maxima.

For item 2, let P be the predictor that, on input $\mathbf{z}$ and with oracle access to H, reads all of H, computes π^i_{H,z_i} — it may, the sources being fixed and predictors being arbitrary, possibly randomised, functions of their inputs, as the statement document's own proof of its fixed-set lemma instructs — and outputs a maximiser. Then $\Pr\left[\mathsf{P}^H(\mathbf{z}) = x_i\right] = \mathbb{E}_{(f,\zeta)}[m_i(f,\zeta)]$, which unpredictability bounds by δ. Item 3 follows from $m_2 \le 1$ and item 2. Item 4 is Cauchy–Schwarz: $\mathbb{E}[\sqrt{m_1} \cdot \sqrt{m_2}] \le \sqrt{\mathbb{E}[m_1]}\sqrt{\mathbb{E}[m_2]} \le \delta$, by item 2 applied to each coordinate separately. Item 5 is Markov applied to the non-negative variable $\sqrt{m_1 m_2}$, together with item 4. □

Item 1 is where the second source is spent: a *single cell* of $[N]^2$ carries mass $m_1 m_2$, a product of two independently small numbers, whereas one unpredictable source over $[N]^2$ gives only δ per cell.

Remark 1 ($\mathbb{E}[m_1 m_2] \le \delta^2$ is false). This matters, because it is what forces the exponent $1/3$ in Lemma 8. Let $2 \le M \le N$, $\delta := 1/M$, let E be the event $f(1,1) = 1$, of probability δ, and let both sources output $x_i := 1$ if E holds and x_i uniform on $[N]$ otherwise, leaking the empty string. The coins are independent, so the pair is split, and $\mathbb{E}[m_i] = \delta + (1-\delta)/N \le 2\delta$, so the pair is 2δ-unpredictable; yet $m_1 m_2 = 1$ on E, so $\mathbb{E}[m_1 m_2] \ge \delta$. Splitness constrains the coins, not the two sources' common dependence on f.

Remark 2 (the second source, spent twice). Item 4 of Lemma 2 does *not* follow from item 3 by Jensen, which runs the other way, and it is a factor $\sqrt{\delta}$ stronger than the $\sqrt{\mathbb{E}[m_1 m_2]} \le \sqrt{\delta}$ that concavity gives. It is where the second source's independence is spent a second time, the first being item 1, and it is what turns the union-bound budget of Lemma 6 into $\delta\sqrt{M}$ rather than $\sqrt{M\delta}$ — the difference between a useful bound and a useless one. It is an equality on two structurally different families: on the one in Remark 1, where $m_1 m_2$ has a heavy tail and $\mathbb{E}[\sqrt{m_1 m_2}] = \delta + (1-\delta)/N = \mathbb{E}[m_i]$ exactly, and on the flat family of Proposition 2, where $m_1 = m_2$ is deterministic.

2 Flattening and rectangle discrepancy

Lemma 3 (flattening). *Let π be a distribution on $[N]$ with $\|\pi\|_\infty = m$ and $k := \lfloor 1/m \rfloor$. Then $1 \le k \le N$, $1/k \le 2m$, and π is a convex combination of distributions each uniform on a subset of size exactly k. Consequently, for every $G : [N]^2 \to [-1,1]$ and every π^1,π^2 with $\|\pi^i\|_\infty = m_i$, $k_i := \lfloor 1/m_i \rfloor$,*

$$\left|\mathbb{E}_{\pi^1\otimes\pi^2}[G]\right| \;\le\; \max\left\{ \left|\mathbb{E}_{\mathrm{Unif}(B_1)\otimes\mathrm{Unif}(B_2)}[G]\right| \;:\; |B_i| = k_i \right\}.$$

Proof. $m \ge 1/N$ gives $k \le N$; $m \le 1$ gives $k \ge 1$. For $x \ge 1$, $\lfloor x \rfloor \ge x/2$ (for $1 \le x < 2$ because $\lfloor x \rfloor = 1 \ge x/2$; for $x \ge 2$ because $\lfloor x \rfloor > x-1 \ge x/2$), so with $x = 1/m$, $1/k \le 2m$. If $k = 1$ the claimed decomposition is the identity $\pi = \sum_u \pi(u)\delta_u$, a convex combination of distributions uniform on singletons, and no external result is needed; assume $k \ge 2$. Since $\|\pi\|_\infty \le 1/k = 2^{-\log k}$, $\log k > 0$ and $2^{\log k} = k \in \mathbb{N}$, the flat-decomposition lemma [CFHS, Lemma 3] applies and writes π as a convex combination of distributions uniform on k-subsets. The displayed inequality follows because $(\rho^1,\rho^2) \mapsto \mathbb{E}_{\rho^1\otimes\rho^2}[G]$ is bilinear, so substituting both decompositions expresses the left side as a convex combination of the quantities on the right. □

Note G is arbitrary: it will carry an indicator $\mathbb{1}[\mathbf{u} \notin \mathcal{S}]$ that does *not* factor over the coordinates. That is harmless, because the linear structure used lives in (π^1,π^2), not in G.

Both concentration bounds below are one Hoeffding estimate with two different union bounds, and both are stated through a single quantity. For $c > 0$ put $L := \ln(eN)$ and

$$t_c(k_1,k_2) := \sqrt{\frac{k_1 \ln \frac{eN}{k_1} + k_2 \ln \frac{eN}{k_2} + c}{2k_1k_2}},$$

where c is the logarithm of the number of tuples the union runs over.

Lemma 4 (the expectation of t_c). *For every $c > 0$, pointwise at $k_i = \lfloor 1/m_i \rfloor$, $t_c(k_1,k_2)^2 \le L(m_1+m_2) + 2c\, m_1 m_2$, and therefore*

$$\mathbb{E}\Big[t_c(k_1,k_2)\Big] \;\le\; \min\left\{ \sqrt{2\delta(L+c)}\;,\;\; \sqrt{2L\delta} + \delta\sqrt{2c} \right\}.$$

Proof. Expanding, $t_c^2 = \frac{\ln(eN/k_1)}{2k_2} + \frac{\ln(eN/k_2)}{2k_1} + \frac{c}{2k_1k_2}$; each $k_i \ge 1$ gives $\ln(eN/k_i) \le L$, and $1/k_i \le 2m_i$ from Lemma 3 gives $\frac{1}{2k_i} \le m_i$ and $\frac{1}{2k_1k_2} \le 2m_1m_2$, which is the pointwise bound. For the first arm, concavity of $\sqrt{\cdot}$ applied to the whole expression with Lemma 2(2),(3) gives $\mathbb{E}[t_c] \le \sqrt{L(\mathbb{E}m_1 + \mathbb{E}m_2) + 2c\mathbb{E}[m_1m_2]} \le \sqrt{2\delta(L+c)}$. For the second, $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ splits the pointwise bound first; concavity and Lemma 2(2) give $\mathbb{E}[\sqrt{L(m_1+m_2)}] \le \sqrt{2L\delta}$, and Lemma 2(4) gives $\sqrt{2c}\,\mathbb{E}[\sqrt{m_1m_2}] \le \delta\sqrt{2c}$. □

Splitting before averaging is what the second arm buys: it pays $\delta\sqrt{2c}$ where the first pays $\sqrt{2c\delta}$. At $c = C_0$ below this is a sharpening of a constant; at $c = C_1$, where c carries the alphabet size, it decides whether the bound says anything.

Definition 1 (revealing rule). A *revealing rule of budget s* is a family $(\mathcal{A}_\zeta)_{\zeta\in\mathcal{Z}}$ of procedures, where $\mathcal{A}_\zeta$ inspects the coordinates of f one at a time — the next a function of ζ and the values already seen — halts having inspected a set $\mathcal{S}(f,\zeta)$ with $|\mathcal{S}(f,\zeta)| \le s$, and outputs a test $\theta_{\zeta,f} : [M] \to \{0,1\}$ that is a function of ζ and the inspected values alone.

Lemma 5 (rectangle discrepancy against leakage-named tests). *Let $\gamma_0 \in (0,1)$ and let $(\mathcal{A}_\zeta)$ be a revealing rule. Put $C_0 := (\sigma+2)\ln 2 + \ln\frac{4N^2}{\gamma_0}$. Then with probability at least $1-\gamma_0$ over $f \leftarrow \mathsf{Fun}$, simultaneously for every $\zeta \in \mathcal{Z}$, all $k_1,k_2 \in [N]$ and all $B_i \subseteq [N]$ with $|B_i| = k_i$, writing $R := B_1 \times B_2$,*

$$\Bigg|\frac{1}{k_1k_2}\sum_{u\in R\setminus\mathcal{S}(f,\zeta)}\big(\theta_{\zeta,f}(f(u)) - p_{\theta_{\zeta,f}}\big)\Bigg| \;\le\; t_{C_0}(k_1,k_2). \tag{1}$$

Proof. Conditional uniformity off the revealed set. Fix ζ. For $S_0 \subseteq [N]^2$ and $w \in [M]^{S_0}$ the event $\mathcal{T}_{S_0,w} := \{\mathcal{S}(f,\zeta) = S_0 \wedge f|_{S_0} = w\}$ is measurable with respect to $f|_{S_0}$: replaying $\mathcal{A}_\zeta$ on w reproduces its whole inspection sequence. The coordinates of f being i.i.d. uniform, conditioning on such an event leaves $f|_{[N]^2\setminus S_0}$ i.i.d. uniform; and $\theta_{\zeta,f}$, $p_{\theta_{\zeta,f}}$ are constant on $\mathcal{T}_{S_0,w}$.

One tuple. Fix also k_1,k_2,B_1,B_2 and condition on $\mathcal{T}_{S_0,w}$. Write θ, p_θ for the constants and $n_0 := |R\setminus S_0| \le k_1k_2$. The variables $\theta(f(u)) - p_\theta$, $u \in R\setminus S_0$, are independent, lie in an interval of length 1, and have mean 0, since $f(u)$ is uniform on $[M]$ and $\Pr\big[\theta(f(u)) = 1\big] = p_\theta$ exactly. Hoeffding's inequality with $\lambda := k_1k_2\, t_{C_0}(k_1,k_2)$ gives conditional failure probability at most $2\exp(-2\lambda^2/n_0) \le 2\exp(-2t_{C_0}^2 k_1k_2)$, uniformly in (S_0,w), hence also unconditionally.

Union bound. By the definition of t_c, $2\exp(-2t_{C_0}^2 k_1k_2) = 2\,e^{-k_1\ln\frac{eN}{k_1}}\,e^{-k_2\ln\frac{eN}{k_2}}\,2^{-(\sigma+2)}\,\frac{\gamma_0}{4N^2}$. There are $|\mathcal{Z}|\binom{N}{k_1}\binom{N}{k_2}$ tuples with the given (k_1,k_2), and $\binom{N}{k} \le (eN/k)^k$ [CFHS, Lemma 4.3] while $|\mathcal{Z}| < 2^{\sigma+2}$. Multiplying, all three exponential factors cancel and the contribution of the pair (k_1,k_2) is at most $2\gamma_0/(4N^2) = \gamma_0/(2N^2)$; summing over the N^2 pairs gives $\gamma_0/2 \le \gamma_0$. □

Lemma 6 (rectangle discrepancy against *all* tests). *Let $\gamma_0 \in (0,1)$ and put $C_1 := M\ln 2 + \ln\frac{4N^2}{\gamma_0}$. Then with probability at least $1-\gamma_0$ over $f \leftarrow \mathsf{Fun}$, simultaneously for all $k_1,k_2 \in [N]$, all $B_i \subseteq [N]$ with $|B_i| = k_i$, and* every *function $\theta : [M] \to \{0,1\}$, writing $R := B_1 \times B_2$,*

$$\Bigg|\frac{1}{k_1k_2}\sum_{\mathbf{u}\in R}\big(\theta(f(\mathbf{u})) - p_\theta\big)\Bigg| \;\le\; t_{C_1}(k_1,k_2).$$

Equivalently, every rectangle's pushforward $\nu_R := f_*\mathrm{Unif}(R)$ satisfies $\mathrm{SD}\big(\nu_R, \mathrm{Unif}([M])\big) \le t_{C_1}(|B_1|,|B_2|)$.

Proof. Fix k_1, k_2, sets B_i with $|B_i| = k_i$, and a function θ — all fixed before f is drawn. The variables $\theta(f(\mathbf{u})) - p_\theta$, $\mathbf{u} \in R$, are independent, the coordinates of f being i.i.d. uniform; each has mean exactly 0, since $\Pr\big[\theta(f(\mathbf{u})) = 1\big] = |\theta^{-1}(1)|/M = p_\theta$; and each lies in $\{-p_\theta, 1 - p_\theta\}$, an interval of length 1. Hoeffding with $\lambda := k_1 k_2\, t_{C_1}$ gives failure probability at most $2\exp(-2t_{C_1}^2\, k_1 k_2)$, which by the definition of t_c equals $2(eN/k_1)^{-k_1}(eN/k_2)^{-k_2}2^{-M}\gamma_0/(4N^2)$. For the given (k_1, k_2) there are $\binom{N}{k_1}\binom{N}{k_2}2^M$ tuples, since there are exactly 2^M functions $[M] \to \{0,1\}$, and $\binom{N}{k} \le (eN/k)^k$ [CFHS, Lemma 4.3]; all three exponential factors cancel, leaving $\gamma_0/(2N^2)$ per pair and $\gamma_0/2 \le \gamma_0$ over the N^2 pairs. For the reformulation, every subset of $[M]$ is $\theta^{-1}(1)$ for exactly one θ, and the displayed quantity is $\big|\nu_R(\theta^{-1}(1)) - \mathrm{Unif}([M])(\theta^{-1}(1))\big|$. □

Neither a conditioning nor a revealed set appears. The bound holds for every θ *fixed in advance*, and the test the observer induces — which does depend on f — may then be substituted, because a statement "for all θ, $\Phi(f,\theta) \le t$" may be instantiated at any θ whatsoever. That single move dispenses with Definition 1, Lemma 7 and Lemma 8 altogether, at the price of $M \ln 2$ inside C_1 where Lemma 5 has $(\sigma + 2)\ln 2$: the leakage leaves the budget and the alphabet takes its place.

Remark 3 (the two routes, and where one source would break both). Lemmas 5 and 6 are the same Hoeffding estimate with two answers to one question: how is a test that depends on f prevented from conspiring with the very values it is tested against? The first pins the test down by revealing the cells that determine it, paying $\Pi(\mathcal{S})$ for those cells and only $|\mathcal{Z}| < 2^{\sigma+2}$ for the union; the second declines to pin it down at all, paying 2^M for the union and nothing for revealed cells. Neither dominates, and §4 carries both. Lemma 6 is not lossy for what it bounds: by its reformulation the supremum over θ is a statistical distance, whose expectation for random f is of order $\sqrt{M/(k_1k_2)}$ when $M \le k_1k_2$, while $t_{C_1} \ge \sqrt{M\ln 2/(2k_1k_2)}$; Proposition 2 makes this a proof. So no smaller test class, chaining argument or sharper tail bound can improve it.

The statement document demands that any proof fail visibly when the two sources are replaced by one. Both routes do, and not in the size of t: at $k_1 = 1, k_2 = N$ one gets $t_{C_0}(1,N)^2 = (L + N + C_0)/(2N) \to \frac{1}{2}$, so $t_{C_0} \to 1/\sqrt{2}$, neither large nor vacuous — and in any case $k_1 = 1$ models a *deterministic first coordinate*, not a single source. The failure is earlier and total. A single source emits a point of $[N]^2$ whose posterior given the oracle and the leakage need not factor at all; Lemma 2(1) is then unavailable, a cell carries mass up to δ rather than m_1m_2, and there is no product measure for Lemma 3 to flatten onto a rectangle. The union would then range over $\binom{N^2}{k}$ arbitrary k-subsets rather than over rectangles, forcing $t^2 \ge \frac{1}{2}\ln(eN^2/k) \ge \frac{1}{2}$ and a vacuous conclusion; and Lemma 2(4), whose Cauchy–Schwarz step is applied across the two coordinates, has nothing to be applied to. Lemmas 3, 5 and 6 have no input and the argument does not begin.

3 The observer as a revealing rule

Definition 2 (challenge resolution). D has *challenge resolution* M' if there is $h : [M] \to [M']$ such that for **every coin string** ρ **of** D, every f, every ζ, and all v, v' with $h(v) = h(v')$, the runs $D^f_\rho(v,\zeta)$ and $D^f_\rho(v',\zeta)$ issue the same sequence of query *positions*; the output bit is unconstrained. Every D has resolution M; resolution 1 means that, for each fixed coin string, the query positions ignore the challenge value. We may take h surjective without loss: replacing $[M']$ by $\mathrm{image}(h)$ issues fewer runs below and only shrinks the budget.

The coin quantifier is part of the definition, not a technicality. The weaker condition that the query positions be challenge-independent *in law* defines a strictly larger class: at $N = M = 2$, $q = 1$, the observer that flips ρ and queries $(1,1)$ when $v \oplus \rho = 1$ and $(2,2)$ otherwise has query position uniform on $\{(1,1),(2,2)\}$ for both challenge values, yet each of its two coin fixings has resolution 2. Such an observer steers its probes by the challenge once its coins are fixed — exactly the behaviour Barrier 1 leaves open — and is not in the class κ^{na} ranges over.

Lemma 7 (the observer as a revealing rule). *Let D be deterministic, q-query, challenge-oblivious, of resolution M'. Then the procedures "$\mathcal{A}_\zeta$: for $j = 1,\dots,M'$ in order, run $D^f(v_j,\zeta)$ for a fixed $v_j \in h^{-1}(j)$ — which exists, h being surjective — inspecting each queried cell not already known" form a revealing rule of budget $s = \min(qM', N^2)$, with $\theta_{\zeta,f}(v) := D^f(v,\zeta)$.*

Proof. $\mathcal{A}_\zeta$ inspects one coordinate at a time, the next being a function of ζ and the values already seen: the loop and the representatives are fixed in advance, and inside run j the next position is a function of (v_j,ζ) and previous answers. Each of the M' runs issues at most q positions, so at most qM' cells, and at most N^2 exist. Finally, for arbitrary v with $j = h(v)$, the run $D^f(v,\zeta)$ issues exactly the positions run j issued, all inspected and recorded, so its whole execution — hence its output bit — is reconstructible from $(v,\zeta,\mathcal{S},f|_{\mathcal{S}})$. Therefore the entire function $\theta_{\zeta,f}$, and $p_{\theta_{\zeta,f}}$, is determined by $(\zeta,\mathcal{S},f|_{\mathcal{S}})$. □

The resolution parameter is the whole difference between the settled and the open case. Lemma 5 needs the test *as a function on all of* $[M]$, because the reference p_θ averages over the alphabet; so every value the observer might receive must have its query path revealed in advance, and an observer that steers by v forces $M' = M$.

Lemma 8 (product mass of the revealed set). *For a revealing rule of budget $s \geq 1$, $\mathbb{E}\big[\min(\Pi_{f,\zeta}(\mathcal{S}(f,\zeta)),1)\big] \leq \mu(s) := \min\big(s\delta,\ 2(s\delta^2)^{1/3},\ 1\big)$; for $s = 0$ the left side is* 0.

Proof. By Lemma 2(1), $\Pi(\mathcal{S}) = \sum_{\mathbf{u}\in\mathcal{S}} \pi^1(u_1)\pi^2(u_2) \leq s\, m_1 m_2$. Taking expectations and using Lemma 2(3) gives the arm $s\delta$; the arm 1 is trivial. For the third, fix $\vartheta \in (0,1]$, bound by $s\vartheta$ on $\{m_1 m_2 \leq \vartheta\}$ and by 1 off it, and use Lemma 2(5): $\mathbb{E}[\min(\Pi(\mathcal{S}),1)] \leq g(\vartheta) := s\vartheta + \delta\vartheta^{-1/2}$. Then $g'(\vartheta) = s - \frac{1}{2}\delta\vartheta^{-3/2}$ vanishes at $\vartheta^* = (\delta/2s)^{2/3}$, which is ≤ 1 since $\delta \leq 1 \leq 2s$, and $g'' > 0$; and $g(\vartheta^*) = \big(2^{-2/3} + 2^{1/3}\big)(s\delta^2)^{1/3} \leq 2(s\delta^2)^{1/3}$. □

By Remark 1 the bound $s\delta^2$ is *not* available, and the exponent $1/3$ is exactly what the $\sqrt{\vartheta}$ of Lemma 2(5) produces. Lemma 8 is tight at $s = 1$ for that example.

4 Extraction

Lemma 9. *For deterministic D with associated $\theta_{\zeta,f}$, $\Pr[\mathsf{Real} = 1] = \mathbb{E}_{(f,\zeta)}\big[\mathbb{E}_{\mathbf{x}\sim\Pi_{f,\zeta}}[\theta_{\zeta,f}(f(\mathbf{x}))]\big]$ and $\Pr[\mathsf{Real}_0 = 1] = \mathbb{E}_{(f,\zeta)}[p_{\theta_{\zeta,f}}]$.*

Proof. Condition on $(H,\mathbf{z}) = (f,\zeta)$. In both experiments D's oracle is f, so its output on challenge v is $\theta_{\zeta,f}(v)$. In Real the challenge is $f(\mathbf{x})$ with $\mathbf{x} \sim \Pi_{f,\zeta}$ by Lemma 2; in Real_0 it is uniform and independent, with $\mathbb{E}_{v\leftarrow[M]}[\theta(v)] = p_\theta$. □

So Real and Real_0 differ in the challenge alone, the oracle being the true H in both: the whole problem is the one-cell question *is $f(\mathbf{x})$ distributed like a fresh uniform value, to a reader who knows ζ and can probe f q times?*

Proposition 1. *For every $\gamma_0 \in (0,1)$ and every deterministic q-query challenge-oblivious D of resolution M',*

$$\big|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\big| \le \gamma_0 + \min\Big\{\sqrt{2\delta(L + C_0)},\ \sqrt{2L\delta} + \delta\sqrt{2C_0}\Big\} + \mu\big(\min(qM',N^2)\big),$$

with $L = \ln(eN)$ and C_0 as in Lemma 5.

Proof. By Lemma 9 and the triangle inequality the left side is at most $\mathbb{E}[\Delta]$ with $\Delta(f,\zeta) := |\mathbb{E}_\Pi[\theta(f(\mathbf{x})) - p_\theta]|$. Put $G_{f,\zeta}(\mathbf{u}) := (\theta(f(\mathbf{u})) - p_\theta)\mathbb{1}[\mathbf{u} \notin \mathcal{S}(f,\zeta)] \in [-1,1]$, so that, since $|\theta - p_\theta| \le 1$,

$$\Delta(f,\zeta) \ \le\ \big|\mathbb{E}_{\Pi_{f,\zeta}}[G_{f,\zeta}]\big| + \Pi_{f,\zeta}(\mathcal{S}(f,\zeta)),$$

the two terms being an exhaustive split of the challenge cell. Lemma 3 bounds the first by the largest $|\mathbb{E}_{\mathrm{Unif}(B_1)\otimes\mathrm{Unif}(B_2)}[G]|$ over $|B_i| = k_i := \lfloor 1/m_i\rfloor$, and for such a rectangle that quantity is

$$\frac{1}{k_1k_2}\sum_{\mathbf{u}\in R} G_{f,\zeta}(\mathbf{u}) = \frac{1}{k_1k_2}\sum_{\mathbf{u}\in R\setminus\mathcal{S}(f,\zeta)} \big(\theta(f(\mathbf{u})) - p_\theta\big),$$

verbatim the quantity Lemma 5 controls; on the event $\mathcal{E}$ of that lemma it is at most $t_{C_0}(k_1,k_2)$. As $\Delta \le 1$ always, splitting on $\mathcal{E}$ gives $\mathbb{E}[\Delta] \le \Pr[\mathcal{E}^c] + \mathbb{E}[t_{C_0}] + \mathbb{E}[\min(\Pi(\mathcal{S}),1)]$, all terms non-negative so the indicators may be dropped; these are at most γ_0, Lemma 4 at $c = C_0$, and $\mu(s)$ by Lemmas 7 and 8. □

Theorem 1 (extraction, challenge-blind query positions). *Put*

$$\Lambda \ :=\ \min\big\{5\sqrt{\sigma'\delta},\ \ \sqrt{2\delta\ln(eN)} + 4\delta\sqrt{\sigma'}\,\big\}.$$

Then $\kappa^{\mathrm{na}}(q) \le \Lambda + \mu(q)$ for every q, and $\kappa(0) \le \Lambda$ with no restriction on the observer. The second arm of Λ is the smaller once $\delta \le 0.85$, and is smaller by a factor $\Theta(\sqrt{\delta})$ on the $\sqrt{\sigma'}$ term as $\delta \to 0$.

Proof. Both $\kappa^{\mathrm{na}}(q)$ and $\kappa(0)$ are suprema over observers the statement permits to randomise, whereas Proposition 1 is stated for deterministic D; by Lemma 1 the suprema are attained on deterministic observers and D' inherits D's challenge resolution, so restricting to deterministic resolution-1 observers loses nothing. If $\delta = 1$ then $\sqrt{\sigma'\delta} \geq \sqrt{2} > 1$ and there is nothing to prove; else take $\gamma_0 := \delta$ in Proposition 1, so $\log \gamma_0^{-1} \leq \log N$ as $\delta \geq 1/N$. Resolution 1 gives $s \leq q$, so the last term is $\mu(q)$; for $q = 0$, $s = 0$ and every observer has resolution 1 vacuously. It remains to bound γ_0 plus each arm of Lemma 4 by the corresponding arm of Λ.

First arm. $\log \gamma_0^{-1} \leq \log N \leq \sigma'/2$ gives $\sqrt{2\delta(L + C_0)} \leq \sqrt{12\sigma'\delta} \leq 3.47\sqrt{\sigma'\delta}$, as in the numerical estimate of the display, and $\gamma_0 = \delta \leq \sqrt{\sigma'\delta}$ because $\delta \leq 1 \leq \sigma'$; total at most $5\sqrt{\sigma'\delta}$.

Second arm. $2C_0 = 2(\sigma+2)\ln 2 + 2\ln 4 + 4\ln N + 2\ln\delta^{-1} \leq 1.3863\sigma + 5.5452 + 6\ln N$, and $\sigma \leq \sigma'$, $5.5452 \leq 2.7726\sigma'$, $6\ln N \leq 2.0794\sigma'$ — the last two from $\sigma' \geq 2\log N$ and $\sigma' \geq 2$ — give $2C_0 \leq 6.238\sigma'$, so $\delta\sqrt{2C_0} \leq 2.498\,\delta\sqrt{\sigma'}$; and $\gamma_0 = \delta \leq 0.708\,\delta\sqrt{\sigma'}$ since $\sqrt{\sigma'} \geq \sqrt{2}$. The two together are at most $3.21\,\delta\sqrt{\sigma'} \leq 4\delta\sqrt{\sigma'}$. For the comparison, $2\delta\ln(eN) \leq 1.6932\,\sigma'\delta$ gives $\sqrt{2\delta\ln(eN)} \leq 1.302\sqrt{\sigma'\delta}$, so the second arm is at most $(1.302 + 4\sqrt{\delta})\sqrt{\sigma'\delta}$, below $5\sqrt{\sigma'\delta}$ exactly when $4\sqrt{\delta} \leq 3.698$. □

Corollary 1. *If $q^+\delta \leq \sigma'$ then $\kappa^{\mathrm{na}}(q) \leq 6\sqrt{\sigma' q^+\delta}$. Outside that regime $q^+\delta > \sigma' \geq 2$, so $\sqrt{\sigma' q^+\delta} > \sigma' > 1$ and the target bound is vacuous.*

Proof. $\Lambda \leq 5\sqrt{\sigma'\delta} \leq 5\sqrt{\sigma' q^+\delta}$; and $q^+\delta \leq \sigma'$ gives $(q^+\delta)^2 \leq \sigma' q^+\delta$, i.e. $\mu(q) \leq q\delta \leq q^+\delta \leq \sqrt{\sigma' q^+\delta}$. □

This settles the statement's line that queries whose positions ignore the challenge value are almost free at every budget: their cost is the additive $q\delta$, with no multiplier on the leading term and no $\log M$.

Theorem 2 (extraction, unrestricted observers). $\kappa(q) \leq \Lambda + \mu(\min(qM, N^2))$ *for every q, with Λ as in Theorem 1.*

Proof. By Lemma 1 the supremum is attained on deterministic observers; apply Proposition 1 to such a D with $M' = M$ and $\gamma_0 = \delta$, as in Theorem 1. □

Corollary 2. $\kappa(q) \leq 6\sqrt{\sigma' q^+\delta}$ *whenever $q^+\delta \leq \sigma'$ and either $M \leq \sqrt{\sigma'/(q\delta)}$ or $M \leq \sigma'^{3/2}\sqrt{q/\delta}/8$. If $M \leq \sigma'/\sqrt{8\delta}$ then one of the two holds for every $q \geq 1$.*

Proof. $qM\delta \leq \sqrt{\sigma' q^+\delta}$ follows from $qM^2\delta \leq \sigma'$, and $2(qM\delta^2)^{1/3} \leq \sqrt{\sigma' q^+\delta}$ from $8qM\delta^{1/2} \leq \sigma'^{3/2}q^{3/2}$, both using $q \leq q^+$; the leading $\Lambda \leq 5\sqrt{\sigma'\delta} \leq 5\sqrt{\sigma' q^+\delta}$. The first arm fails only if $q > \sigma'/(M^2\delta)$ and the second only if $q < 64M^2\delta/\sigma'^3$; both fail at some q only if $\sigma'^4 < 64M^4\delta^2$, i.e. $M > \sigma'/\sqrt{8\delta}$. □

A bound that does not mention the query budget

Theorem 3 (extraction, unrestricted observers, no query budget). *For every $\gamma_0 \in (0,1)$ and every challenge-oblivious observer D — of any query count, bounded or unbounded, and any challenge resolution —*

$$\left|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\right| \le \gamma_0 + \sqrt{2\delta \ln(eN)} + \delta\sqrt{2M\ln 2 + 2\ln\frac{4N^2}{\gamma_0}}.$$

Consequently $\kappa(q) \le 5\sqrt{\sigma'\delta} + 2\delta\sqrt{M}$ for every q.

Proof. By Lemma 1 we may take D deterministic: that lemma bounds no query count, and nothing here refers to one. By Lemma 9 the left side is at most $\mathbb{E}[\Delta]$ with $\Delta(f,\zeta) = \left|\mathbb{E}_\Pi[\theta_{\zeta,f}(f(\mathbf{x})) - p_{\theta_{\zeta,f}}]\right|$. Fix (f,ζ) and put $G_{f,\zeta}(\mathbf{u}) := \theta_{\zeta,f}(f(\mathbf{u})) - p_{\theta_{\zeta,f}} \in [-1,1]$; by Lemma 2 the law $\Pi_{f,\zeta}$ is a product, so Lemma 3 bounds Δ by the largest $\left|\frac{1}{k_1k_2}\sum_{\mathbf{u}\in R} G_{f,\zeta}(\mathbf{u})\right|$ over $|B_i| = k_i$. On the event of Lemma 6 — a property of f alone, of probability at least $1-\gamma_0$ — that maximum is at most $t_{C_1}(k_1,k_2)$: the pair (k_1,k_2) lies in $[N]\times[N]$ by Lemma 3, and $\theta_{\zeta,f}$, although built from f, is one particular element of the class that lemma quantifies over. Since $\Delta \le 1$ always, $\mathbb{E}[\Delta] \le \gamma_0 + \mathbb{E}[t_{C_1}] \le \gamma_0 + \sqrt{2L\delta} + \delta\sqrt{2C_1}$ by Lemma 4 at $c = C_1$. Nothing above uses the observer's query count, its resolution, or the revealing apparatus of §3.

For the consequence, take $\gamma_0 := \delta$; if $\delta = 1$ then $5\sqrt{\sigma'\delta} > 1$ and there is nothing to prove. As $\delta \ge 1/N$,

$$\begin{aligned} 2C_1 &= 2M\ln 2 + 2\ln 4 + 4\ln N + 2\ln\delta^{-1} \\ &\qquad\qquad \le 1.3863M + 2.7726 + 6\ln N \le 1.3863M + 3.4657\,\sigma', \end{aligned}$$

using $2.7726 \le 1.3863\sigma'$ and $6\ln N \le 2.0794\sigma'$, so $\delta\sqrt{2C_1} \le 1.1774\,\delta\sqrt{M} + 1.8617\,\delta\sqrt{\sigma'}$. Now $\delta\sqrt{\sigma'} = \sqrt{\delta}\sqrt{\sigma'\delta} \le \sqrt{\sigma'\delta}$, $\gamma_0 = \delta \le \sqrt{\sigma'\delta}$, and $\sqrt{2\delta\ln(eN)} \le 1.302\sqrt{\sigma'\delta}$; adding, $(1 + 1.302 + 1.8617)\sqrt{\sigma'\delta} + 1.1774\,\delta\sqrt{M}$. □

For $M = 1$ the set Fun is a singleton, so Real and Real_0 are the same experiment and the advantage is 0; the display is stated for $M \ge 2$ only because C_1 is.

Corollary 3. *Let $q \ge 1$ and $M \ge 4$. If $2\delta\sqrt{M} < 1$ then $2\delta\sqrt{M} \le \mu(\min(qM, N^2))$, so Theorem 3 is the smaller of it and Theorem 2 wherever either says anything at all. Moreover $2\delta\sqrt{M} \le \sqrt{\sigma' q^+\delta}$ precisely when $M \le \sigma' q^+/(4\delta)$.*

Proof. Against μ's arm 1: immediate from $2\delta\sqrt{M} < 1$. Against $2(s\delta^2)^{1/3}$ with $s \le qM$: $2\delta\sqrt{M} \le 2(qM\delta^2)^{1/3}$ iff $\delta\sqrt{M} \le q$, and $\delta\sqrt{M} < 1/2 < q$. Against $s\delta \le qM\delta$: iff $q\sqrt{M} \ge 2$, which holds for $q \ge 1$ and $M \ge 4$. The cap $s = N^2$ never binds, since $\delta \ge 1/N$ gives $N^2\delta \ge N \ge 2$ and $2(N^2\delta^2)^{1/3} = 2(N\delta)^{2/3} \ge 2$, so $\mu(N^2) = 1$. The last claim is $4\delta^2 M \le \sigma' q^+\delta$. □

5 Presampling, and the decomposition

Lemma 10 (presampling, oracle-indexed and consistent). *Let $P \in \mathbb{N}$, $\gamma \in (0,1)$.* (For $M = 1$, Fun is a singleton, so Real_0 and Dec_0 are the same experiment and the bound holds with left side 0; the construction divides by $\log M$ and is stated for $M \geq 2$.) *There is a family $\mathcal{Y}^{P,\gamma} = \{Y_{f,\zeta}\}$ indexed by $f \in$ Fun and $\zeta \in \{0,1\}^* \times \{0,1\}^*$ — with $Y_{f,\zeta}$ uniform on Fun at every ζ outside the support of $\mathbf{z}$, so the family is total, no experiment reaching such a ζ — depending only on (S_1, S_2, P, γ), in which every $Y_{f,\zeta}$ is a P-mixture consistent with f, indeed a single P-bit-fixing source, such that for every q and every q-query observer taking $\mathbf{z}$ and an independent uniform value of $[M]$,*

$$\left|\Pr[\mathsf{Real}_0 = 1] - \Pr[\mathsf{Dec}_0 = 1]\right| \;\leq\; \frac{q(\sigma + 2 + 2\log\gamma^{-1})}{P} + 2\gamma.$$

Proof. Step 1: deficiency under randomised leakage. For ζ with $\Pr[\mathbf{z} = \zeta] > 0$ let μ_ζ be the law of H given $\mathbf{z} = \zeta$ and $S_\zeta := N^2 \log M - H_\infty(\mu_\zeta)$. Since $\mu_\zeta(f) = \Pr\big[H = f\big]\Pr\big[\mathbf{z} = \zeta \mid H = f\big]/\Pr[\mathbf{z} = \zeta] \leq (|\mathsf{Fun}|\Pr[\mathbf{z} = \zeta])^{-1}$, we get $S_\zeta \leq \log(1/\Pr[\mathbf{z} = \zeta])$. With $\bar{S} := \sigma + 2 + \log\gamma^{-1}$ and $\mathcal{B} := \{\zeta : S_\zeta > \bar{S}\}$, every $\zeta \in \mathcal{B}$ has $\Pr[\mathbf{z} = \zeta] < \gamma 2^{-(\sigma+2)}$, so $\Pr[\mathbf{z} \in \mathcal{B}] < |\mathcal{Z}|\gamma 2^{-(\sigma+2)} < \gamma$. This replaces the source's own expectation bound on the deficiency, which assumes the advice is a deterministic function of the oracle and fails here.

Step 2: the family. Fix ζ and set $\delta_\zeta := (S_\zeta + \log\gamma^{-1})/(P\log M)$. If $\delta_\zeta > 1$ then $(S_\zeta + \log\gamma^{-1})/P > \log M \geq 1$, so the asserted bound exceeds $q \geq 1$ and is vacuous for $q \geq 1$, while for $q = 0$ both experiments give the observer a uniform challenge and no oracle access; set $Y_{f,\zeta}$ uniform. (The governing condition $\delta_\zeta > 1$ is q-free, so the family is still defined before q.) Otherwise apply the decomposition [CDGS, Claim 2] to μ_ζ with parameter δ_ζ; since $(S_\zeta + \log\gamma^{-1})/(\delta_\zeta \log M) = P$ it yields $\mu_\zeta = \sum_j \lambda_j X_j + \lambda_{\mathrm{fin}} Y_{\mathrm{fin}}$ with $\lambda_{\mathrm{fin}} \leq \gamma$, each X_j a $(P, 1-\delta_\zeta)$-dense source, all supports pairwise disjoint, and each X_j fixing its set I_j, $|I_j| \leq P$, to values that every f in its support takes. Let Y_j be the corresponding P-bit-fixing source and define $Y_{f,\zeta} := Y_j$ for the unique j with $f \in \operatorname{supp} X_j$, and uniform otherwise. Consistency is immediate; the construction reads only (f,ζ) and the decomposition, which depends only on (S_1, S_2, P, γ). Write $J = J(f,\zeta)$.

Step 3: one component. Fix $\zeta \notin \mathcal{B}$ with $\delta_\zeta \leq 1$ and condition on $J = j$. Because the decomposition is of μ_ζ itself, $\lambda_j X_j$ is the restriction of μ_ζ to $\operatorname{supp} X_j$, so the conditional law of H is exactly X_j; in Dec_0 the oracle is drawn freshly from Y_j. The challenge is uniform and independent, so conditioning on it leaves a fixed q-query adaptive distinguisher between X_j and Y_j; [CDGS, Claim 3] bounds its advantage by $q\delta_\zeta \log M = q(S_\zeta + \log\gamma^{-1})/P$, and averaging over the challenge preserves this.

Step 4: averaging. Averaging over j and bounding the residue event, of conditional probability at most γ, by 1; then averaging over ζ, bounding $\mathcal{B}$'s contribution by $\Pr[\mathbf{z} \in \mathcal{B}] < \gamma$ and using $S_\zeta \leq \bar{S}$ off $\mathcal{B}$, gives the claim. □

Theorem 4. *With $\mathcal{Y} := \mathcal{Y}^{P,\gamma/2}$, for every q and every q-query challenge-oblivious D, writing $\varepsilon(D) := \left|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\right|$ for that same D,*

$$\mathsf{Adv}_{\mathcal{Y},D} \;\leq\; \varepsilon(D) + \frac{2(\sigma' + \log\gamma^{-1})q}{P} + \gamma + P\delta + q\delta.$$

Since $\varepsilon(D) \le \kappa(q)$, the same holds with $\kappa(q)$ in place of $\varepsilon(D)$; but the D-relative form is what Theorem 5 uses, and the inequality $\kappa(q) \le \kappa^{\mathrm{na}}(q)$ is false in general and nowhere needed.

Proof. Interpolate $\mathsf{G}_0 = \mathsf{Real}$, $\mathsf{G}_1 = \mathsf{Real}_0$, $\mathsf{G}_2 = \mathsf{Dec}_0$, $\mathsf{G}_3 = \mathsf{Dec}$.

$|\mathsf{G}_0 - \mathsf{G}_1| = \varepsilon(D)$ by definition, for the fixed D the theorem is instantiated at; no supremum is taken here.

$|\mathsf{G}_1 - \mathsf{G}_2|$: Lemma 10 with slack $\gamma/2$ gives $q(\sigma + 2 + 2\log(2/\gamma))/P + \gamma$; as $N \ge 2$ gives $\sigma + 2 \le \sigma'$ and $\sigma' \ge 2$, the numerator is at most $2\sigma' + 2\log\gamma^{-1}$. For $q \ge P$ that term already exceeds $2\sigma' > 1$, so no hypothesis $q < P$ is needed.

$|\mathsf{G}_2 - \mathsf{G}_3| \le P\delta + q\delta$: draw H, $(\mathbf{x},\mathbf{z})$ and J with fixed set I_J; let $H^\circ \sim Y_J$ and $U \leftarrow [M]$ be independent of everything else; let H^* be H° with its value at $\mathbf{x}$ overwritten by U when $\mathbf{x} \notin I_J$. As J is a function of $(H,\mathbf{z})$, H° is independent of $\mathbf{x}$ given $(H,\mathbf{z},J)$; conditioning on $\mathbf{x} = \mathbf{u} \notin I_J$, the source Y_J is uniform and independent at $\mathbf{u}$, so both H° and H^* are distributed as Y_J. Running G_3 with oracle H^* and G_2 with oracle H° and challenge U, the two executions receive the same input on $\{\mathbf{x} \notin I_J\}$, since there $H^*(\mathbf{x}) = U$, and their oracles agree off $\mathbf{x}$. So they diverge only if $\mathbf{x} \in I_J$ or D queries $\mathbf{x}$; being identical up to the step before a first query at $\mathbf{x}$, that event has equal probability in both and may be measured in G_2. The statement document's lemma *the fixed set misses the challenge* gives $\Pr[\mathbf{x} \in I_J] \le P\delta$ — its proof requires only that $\mathcal{Y}$ depend on (S_1, S_2, P, γ) alone, be indexed by (f,ζ), and have its component drawn independently of $\mathbf{x}$ given $(H,\mathbf{z})$, all of which hold, J being a function of $(H,\mathbf{z})$. Its lemma *the observer misses the challenge* gives $\Pr[\mathbf{x} \in Q] \le q\delta$ in G_2, its predictor being simulable there because the challenge is uniform and independent. □

Theorem 5 (the conjecture, on the region reached). *Suppose $q^+\delta \le \sigma'$ and $P \le \sqrt{\sigma'/\delta}$, and suppose at least one of*

1. *$M \le \sigma' q^+/(4\delta)$, with* no *restriction on D;*
2. *D has challenge resolution* 1*;*
3. *the hypothesis of Corollary 2, in particular $M \le \sigma'/\sqrt{8\delta}$.*

Then with $\mathcal{Y} = \mathcal{Y}^{P,\gamma/2}$,

$$\mathsf{Adv}_{\mathcal{Y},D} \;\le\; \frac{2(\sigma' + \log\gamma^{-1})q^+}{P} + 8\sqrt{\sigma' q^+ \delta} + \gamma,$$

that is, the conjecture holds on that region with $c = 2$, $C = 8$.

Proof. $\varepsilon(D) \le 6\sqrt{\sigma' q^+\delta}$ in each branch. Under 1, Theorem 3 gives $\varepsilon(D) \le 5\sqrt{\sigma'\delta} + 2\delta\sqrt{M} \le 5\sqrt{\sigma' q^+\delta} + \sqrt{\sigma' q^+\delta}$, the second term by Corollary 3. Under 2, D has challenge resolution 1, so $\varepsilon(D) \le \kappa^{\mathrm{na}}(q)$ and Corollary 1 applies — equivalently, Proposition 1 bounds $\varepsilon(D)$ directly at $M' = 1$. Under 3, Corollary 2 bounds $\kappa(q) \ge \varepsilon(D)$. Then $P\delta \le \sqrt{\sigma'\delta} \le \sqrt{\sigma' q^+\delta}$ by the hypothesis on P, and $q\delta \le q^+\delta \le \sqrt{\sigma' q^+\delta}$ by $q^+\delta \le \sigma'$. Summing, $6 + 1 + 1 = 8$. □

The family is built from (P,γ) before q and the bound holds for every q, so the statement's quantifier order is respected; and every $Y_{f,\zeta}$ is consistent with f, so the conclusion is the conjecture and not the weakening its remark on consistency describes.

Branch 1 is the one that carries no hypothesis on the observer, and it is what removes from its region the obstruction §8 calls Barrier 1. It contains branch 3's clean form whenever $\delta \le 1/2$, since then $\sigma' q^+/(4\delta) \ge \sigma'/(4\delta) \ge \sigma'/\sqrt{8\delta}$, and enlarges it by the factor $\sqrt{8}\, q^+/(4\sqrt{\delta})$, which is large exactly when δ is small — the regime the conjecture is about. Branch 2 is *not* subsumed: it covers every M, for that restricted class of observers alone. What is known is the union of the three.

6 Combination with the compression bound

For $q \ge 1$, [CFHS, Theorem 3] with $\ell = 2$, $2^{-k} = \delta$, $q_D = q$ gives $\kappa(q) = O\big(\log M \cdot (q\delta^2(\sigma + 2N))^{1/3}\big)$. Hence

$$\kappa(q) \le 5\sqrt{\sigma'\delta} + \min\Big\{\mu(\min(qM, N^2)),\ O\big(\log M\, (q\delta^2(\sigma + 2N))^{1/3}\big)\Big\},$$

and Theorem 4 converts either arm. The arms are complementary: Theorem 2 is smaller when $M \lesssim N \log^3 M$, the compression arm when M is very large relative to N; the latter also relaxes as q grows, giving the target shape with no condition on M once $q \gtrsim \log^6 M (\sigma + 2N)^2 \delta/\sigma'^3$. What neither arm reaches is M large together with q small. Recorded against that source: its right-hand side vanishes at $q_D = 0$ whereas $\kappa(0) > 0$ in general, so it is invoked only for $q \ge 1$, and its constant is not extracted, so this arm is asymptotic while everything else here is explicit.

7 The alphabet term is tight

Proposition 2. *Let $N \ge 2$ and $2 \le M \le N^2/2$. Let S_1 and S_2 each draw $x_i \leftarrow [N]$ on their own private coins and output (x_i, ε). Then the pair is split, $\sigma_1 = \sigma_2 = 0$, and it is δ-unpredictable with $\delta = 1/N$; and there is a deterministic challenge-oblivious observer D^*, making N^2 queries and of challenge resolution* 1, *with*

$$\big|\Pr[\mathsf{Real} = 1] - \Pr[\mathsf{Real}_0 = 1]\big| \ \ge\ \frac{\delta\sqrt{M}}{4\sqrt{2}} \ =\ \frac{\sqrt{M}}{4\sqrt{2}\,N}\,.$$

Proof. Parameters. The coins are independent, so the pair is split. Here $\mathbf{z} = (\varepsilon, \varepsilon)$ always, and each x_i is uniform on $[N]$ and independent of H and of x_{3-i}. A predictor is an arbitrary, possibly randomised function of $(\mathbf{z}, H)$, so its output is independent of x_i and $\Pr\big[\mathsf{P}^H(\mathbf{z}) = x_i\big] = 1/N$ for every P: the pair is δ-unpredictable exactly for $\delta = 1/N$, the least value the statement permits.

The observer. D^* queries all N^2 cells, so holds f; it forms $\nu(v) := |f^{-1}(v)|/N^2$ and on challenge v outputs $\mathbb{1}[\nu(v) > 1/M]$. It is challenge-oblivious, deterministic, makes N^2 queries, and issues the same query positions for every challenge value, so its resolution is 1.

The advantage is an expected statistical distance. Condition on $H = f$. In Real the point $\mathbf{x}$ is uniform on $[N]^2$ and independent of f, so the challenge $f(\mathbf{x})$ has law exactly ν; in Real_0 it is uniform on $[M]$. Hence

$$\Pr[\mathsf{Real} = 1 \mid f] - \Pr[\mathsf{Real}_0 = 1 \mid f] = \sum_{v\in[M]} \left(\nu(v) - \tfrac{1}{M}\right)^+ = \mathrm{SD}(\nu, \mathrm{Unif}([M])) \;\geq\; 0,$$

so no cancellation occurs on averaging and the advantage is $\mathbb{E}_f[\mathrm{SD}(\nu, \mathrm{Unif}([M]))]$.

A fourth-moment lower bound. Let $Y = \sum_{i=1}^n Y_i$ with Y_i i.i.d., $\mathbb{E}Y_1 = 0$, $|Y_1| \leq 1$, and $s^2 := \mathbb{E}Y^2 = n\mathbb{E}Y_1^2$; we claim $\mathbb{E}|Y| \geq s/2$ whenever $s \geq 1$. Discarding the terms carrying an index to an odd power, which vanish by independence and centring, $\mathbb{E}Y^4 = n\mathbb{E}Y_1^4 + 3n(n-1)(\mathbb{E}Y_1^2)^2 \leq n\mathbb{E}Y_1^2 + 3n^2(\mathbb{E}Y_1^2)^2 = s^2 + 3s^4$, using $\mathbb{E}Y_1^4 \leq \mathbb{E}Y_1^2$, valid as $|Y_1| \leq 1$. Hölder with exponents $3/2$ and 3 gives $s^2 = \mathbb{E}[|Y|^{2/3}|Y|^{4/3}] \leq (\mathbb{E}|Y|)^{2/3}(\mathbb{E}Y^4)^{1/3}$, so $\mathbb{E}|Y| \geq s^3/\sqrt{s^2+3s^4} \geq s^3/(2s^2) = s/2$ for $s \geq 1$.

Assembling. Put $n := N^2$, $p := 1/M$ and $X_v := |f^{-1}(v)|$. Over uniform f the indicators $\mathbb{1}[f(\mathbf{u}) = v]$ are i.i.d. Bernoulli(p), so $X_v \sim \mathrm{Bin}(n, p)$ and each centred indicator lies in $[-p, 1-p]$, of absolute value at most 1. Since $\mathrm{SD}(\nu, \mathrm{Unif}([M])) = \frac{1}{2n}\sum_v |X_v - np|$ and the X_v are identically distributed, $\mathbb{E}_f[\mathrm{SD}] = \frac{M}{2n}\mathbb{E}|X - np|$. Here $s^2 = np(1-p) = \frac{N^2}{M}(1 - \frac{1}{M}) \geq \frac{N^2}{2M}$ as $M \geq 2$, and $s^2 \geq 1$ as $M \leq N^2/2$; so $\mathbb{E}|X - np| \geq \frac{N}{2\sqrt{2M}}$ and $\mathbb{E}_f[\mathrm{SD}] \geq \frac{M}{2N^2} \cdot \frac{N}{2\sqrt{2M}} = \frac{\sqrt{M}}{4\sqrt{2}N}$. □

Three things follow. First, Theorem 3 is tight in M to within a factor $8\sqrt{2} < 12$, so **no bound free of q can improve its alphabet term** and Remark 3's claim that Lemma 6 is not lossy is proved. Second, D^* has resolution 1, so the lower bound applies to κ^{na} as much as to κ: whatever remains at large M is *not* the phenomenon Barrier 1 is named for. Consistently, Theorem 1 says nothing here, $\mu(N^2)$ being 1. Third, at $M = \lfloor N^2/2 \rfloor$ the advantage is at least $\frac{1}{8}$ while $\sqrt{\sigma'\delta} = \sqrt{2\log N/N} \to 0$, so no bound $\kappa(q) \leq h(\sigma', \delta, N)$ free of both M and q can hold: the statement's q^+ is necessary. Nothing is refuted — at those parameters the target $C\sqrt{\sigma' q^+ \delta} \geq C\sqrt{2N\log N}$ is itself vacuous.

8 What is not proved

Barrier 1: challenge-steered probes, at M beyond $\sigma' q^+/(4\delta)$. Uncovered: observers of resolution above 1 at $M > \sigma' q^+/(4\delta)$, with q in the window left by Corollary 2 and the compression arm — that is, M large together with q *small.* The mechanism on the revealing route is unchanged and is not slack in the counting. Lemma 5 compares $\mathbb{E}_\Pi[\theta(f(\mathbf{x}))]$ with p_θ, an average over *all* of $[M]$, so θ must be known everywhere; by Lemma 7 an observer that steers its probes by the challenge makes $\theta(v)$ depend on q cells varying with v, so all qM must be revealed before the test is fixed; and Lemma 8 is then tight for what it is given, because coordinatewise δ-unpredictability does not yield $\mathbb{E}[m_1 m_2] \leq \delta^2$ (Remark 1). An observer whose query positions are challenge-independent only *in law*, steering once its coins are fixed, has per-coin resolution above 1 and sits here rather than in Theorem 5's branch 2.

What Lemma 6 does is decline that route entirely, and it is enough for every M up to $\sigma' q^+/(4\delta)$. Beyond that it stops, and Proposition 2 shows it stops for a reason: its alphabet term is within a factor 12 of the truth, so the residue cannot be reached by any q-free estimate. The sharp question is therefore whether $\kappa(q) \le O(\sqrt{\sigma'\delta}) + \mu(q)$ holds for unrestricted observers — whether the revealing route's arm can be made M-free. Both arms in hand are individually tight, μ at $s = 1$ on Remark 1's family and $2\delta\sqrt{M}$ on Proposition 2's, and nothing here claims their minimum is. Two exits on the revealing route, neither taken: replace the reference p_θ by $\theta(f'(\mathbf{x}))$ for an independent copy f' of f off a single cell, cutting the budget from qM to q at the price of proving Lemma 5 for a random, f-dependent reference; or bound $\Pi(S)$ for *reachable* query sets — unions of M decision-tree paths — rather than arbitrary qM-sets. An interpolation between the two routes, revealing the runs for v in a subset $T \subseteq [M]$ and union-bounding over the $2^{M-|T|}$ completions, costs $\mu(q|T|) + \delta\sqrt{2(C_0 + (M - |T|)\ln 2)}$ and yields nothing beyond the endpoints: driving the first term below 1 forces $|T| \lesssim 1/(q\delta^2)$, leaving $M - |T| = M(1 - o(1))$ whenever $\delta\sqrt{M} \gtrsim 1$.

Barrier 2: large P with growing q. Theorem 4 loses $P\delta$ at $\mathsf{G}_2 \to \mathsf{G}_3$, capping the range at $P \le \sqrt{\sigma'/\delta}$; the statement document names this as a separate open problem and nothing here improves it. The slack is identifiable: on $\{\mathbf{x} \in I_J\}$, which is what costs $P\delta$, consistency makes the decomposed challenge $H^*(\mathbf{x})$ equal the real challenge $H(\mathbf{x})$, so the event does not harm $\mathsf{Adv}_{\mathcal{Y},D}$ at all — only the route through the uniform-challenge worlds. A proof comparing G_3 with G_0 directly would not pay it.

9 External results used

- **[CDGS, Claim 2]** (Coretti, Dodis, Guo, Steinberger, *Random Oracles and Non-Uniformity*, ePrint 2017/937, App. A). *A distribution on $[M]^n$ with min-entropy deficiency S is γ-close to a convex combination of finitely many $(P', 1-\delta)$-dense sources,* $P' = (S + \log 1/\gamma)/(\delta \log M)$. The paper states this for a uniform variable conditioned on a deterministic function; its proof uses only the min-entropy deficiency, and the form above is what that proof establishes. The proof's structure — conditioning on $\{Y_I = y_I\}$ and recursing on $\{Y_I \ne y_I\}$ — also supplies the disjointness of the components' supports, the fact that fixed values are values the sampled f takes, and, via its recursion invariant, that the decomposition is of the conditioned distribution itself, so each component is its restriction and the residue's mass is at most γ measured under it and not under the uniform distribution. All three are used in Lemma 10.

- **[CDGS, Claim 3]** (same paper). *For a $(P', 1-\delta)$-dense source X' and its corresponding P'-bit-fixing Y', and any adaptive T-query distinguisher,* $|\Pr[D^{X'} = 1] - \Pr[D^{Y'} = 1]| \le T\delta \log M$.

- **[CFHS, Lemma 3]** (Coretti, Farshim, Harasser, Southern, *Multi-Source Randomness Extraction and Generation in the ROM*, ePrint 2025/1258). *For $k \in \mathbb{R}_{>0}$ with $2^k \in \mathbb{N}$, every k-source is a convex combination of flat k-sources.*

- **[CFHS, Lemma 4.3]** $(n/k)^k \le \binom{n}{k} \le (en/k)^k$.
- **[CFHS, Theorem 3]** *For ℓ sources each $(N^\ell, 2^{-k})$-unpredictable with unbounded oracle access and a q_D-query distinguisher,* $\mathsf{Adv}^{\mathrm{mse}} = O(\ell \log M \sqrt[\ell+1]{q_D 2^{-k\ell}(\sigma + \ell N)})$. Used only for the compression arm.
- **Hoeffding's inequality**, two-sided, for independent variables with ranges of length c_i.
- **Cauchy–Schwarz, Markov, Jensen and Hölder**, the last at exponents $3/2$ and 3, in Lemmas 2, 4, 8 and Proposition 2.

10 Review status

This document has **not** been independently verified, and a reader should weigh it accordingly.

Two rounds of adversarial review were run, six passes recording a verdict and two aborting on infrastructure. **No pass returned clean**. Five findings were upheld across the two rounds, of which two were serious; both were repaired, and both were of the same species — a claim left underdetermined rather than false, each recoverable from machinery already present. Zero numbered conclusions were falsified at any point.

Three limitations of that review are material. Every referee, the adjudicator and the author are the same vendor's models, so the errors that survive are the correlated ones by construction; both serious findings came from the strongest model and were missed by weaker ones examining the same lines. No pass satisfied the campaign's own blindness condition, which requires a process carrying none of the project context and was unavailable in this environment. And no protocol document existed, so every referee ran on instructions written by the session that produced the proof.

The text above is the third revision, extended. Its own tally is empty: nothing in it has been reviewed, the repair that produced the third revision touched Definition 2, on which every resolution argument depends, and the extension — Lemma 2(4),(5), Lemma 4, Lemma 6, Theorem 3, Corollary 3, branch 1 of Theorem 5 and Proposition 2 — has been through no verification pass. One in-session smell test, which is explicitly not such a pass and recorded no verdict in any tally, raised two items, both repaired here and neither load-bearing: Lemma 1 conjoined the two paired experiments where its proof supplies a maximiser per pair, and the guide's aside on (D) claimed of a collected bound what holds only of a display. Its load-bearing new step is a single application of Cauchy–Schwarz, Lemma 2(4); a reader with time for one check should spend it there and on the substitution of $\theta_{\zeta,f}$ into Lemma 6, where the quantifier order does all the work.